

Siddharth Parakh

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🔗 dufka54.github.io

SUMMARY

Analytical Computer & Communication Engineering undergraduate specializing in Machine Learning and Deep Learning architectures. Proficient in PyTorch, Python, and SQL for developing data-driven predictive systems, custom neural network components, and automated data pipelines.

EDUCATION

Manipal University Jaipur 08/2023 – 05/2027 (Expected) | Jaipur, Rajasthan
Bachelor of Technology - Computer and Communication Engineering
Overall CGPA: 8.43

Masai School, Minor in Artificial Intelligence and Data Science, IIT Mandi 06/2025 – 05/2026 | Distance Learning
Overall CGPA: 7.75

SKILLS

Languages (Python, SQL, C++, GDScript, HTML, CSS) | **Libraries** (Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch) |
Tools (Jupyter Notebook, MySQL, Google Colab, Streamlit, VS Code)

PROJECTS

MNIST Translation Invariance Analysis: FNN vs. CNN Performance - PyTorch | Matplotlib

- Designed and trained a Feedforward Neural Network (FNN) and a Convolutional Neural Network (CNN) from scratch using **PyTorch** to evaluate structural variations in digit classification on the MNIST dataset.
- Leveraged convolutional localized feature extraction and spatial weight-sharing to design a progressive filter model (1 -> 16 -> 32 -> 64) that achieved a **99.31% peak validation accuracy using only 46,570 trainable parameters** — delivering a nearly **10x computational footprint reduction** over the rigid 476,490-parameter FNN.
- Developed a custom data pipeline using **Pandas** and **Torchvision** to apply a 5-pixel spatial right-shift on testing datasets, assessing model resilience against geographic translation and spatial distribution shifts.
- Proved structural invariance advantages under spatial translation; the coordinate-dependent FNN suffered a complete structural collapse with accuracy plummeting from **97.46% to 17.50%**, whereas the CNN retained high residual tolerance at **56.76% accuracy**.
- Github link: [Dufka54/MNIST_Spatial_Inversion](#): Comparative analysis between FNN and CNN. 🔗

Student Placement Predictor - Pandas | Scikit-Learn | Matplotlib

- Engineered a custom Linear Regression framework entirely from scratch utilizing Python and foundational mathematical principles, implementing a Batch Gradient Descent optimization algorithm to predict student placement outcomes (Binary: Placed vs. Not Placed).
- Structured a data pipeline by partitioning datasets into distinct training, validation, and testing sets. Conducted an exhaustive grid search to isolate optimal learning rates and epoch thresholds, effectively mitigating overfitting and maximizing model convergence.
- Integrated a custom classification layer by implementing a 0.5 probability decision threshold to convert continuous linear outputs into discrete binary classifications, yielding a peak predictive test accuracy of 79.25%.
- GitHub link: [Dufka54/ML_firstApp](#) 🔗 | Live Demo: [mlfirstapp-2qivgrlwrddxcqc7gvmt6.streamlit.app](#)

Customer Segmentation - Pandas | Numpy | Matplotlib

- Engineered a custom, scratch-built K-Means clustering algorithm in Python using NumPy vectorization and array broadcasting, supporting multi-dimensional Euclidean distance calculations and early convergence detection.
- Built an end-to-end unsupervised learning pipeline on the Mall Customer dataset, incorporating feature engineering, ordinal binning for spending profiles, and Min-Max spatial normalization.
- Implemented hyperparameter grid search optimization across multiple cluster configurations (\$k\$) and iteration epochs, evaluating model separation and cluster compactness using Silhouette Scores and Within-Cluster Sum of Squares
- GitHub link: [Dufka54/Customer-segmentation-app](#) 🔗 | Live: <https://customer-segmentation-app-7tbickmpdwrxlouxmsjs2v.streamlit.app/> 🔗

CERTIFICATION

Design and Analysis of Algorithm (NPTEL)

Result: 61%

Introduction to Database Systems Certificate 🔗, NPTEL

Result: 69% | Credential ID: NPTEL25CS40S647402071

01/2025 – 04/2025